

Generation Interconnection System Impact Study Report

For Queue Project Q22-2257

Pendleton 100 kV

January 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2257. The Interconnection Customer proposes a 199.1 MW battery energy storage facility interconnecting at the Pendleton 100 kV substation in the Indiana area. The request was received on December 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 433 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,545,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rmec-S Haven 100 kV	South Haven-Napavine 100 kV	174.5	161	99.9
S Haven-Napavine 100 kV Line	South Haven-Rmec 100 kV	151.8	185	99.9
Anderson-Pendleton 100 kV	Trego-Swan Vly 500 kV	149.5	161	100.0
Napavine-Murdock 100 kV	S Haven-Rmec Substation 100 kV	135.4	198	99.9
Rmec-Christopher 100 kV line	Napavine-S Haven 100 kV	129.3	144	72.9
Wrangell-Devers 100 kV	Montpelier Electric Generating Statin-Muskogee Substation 500 kV	102.1	113	9.0
Shiprock 2 Substation-Makahoy 100 kV	Muskogee-Montpelier Electric Generating Sta 500 kV	100.9	120	9.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rmec-Suth Haven 100 kV	Rebuild line with higher-capacity conductor	\$4,193,000	79% (\$3,297,000)
S Haven-Napavine Substation 100 kV	Rebuild line with higher-capacity conductor	\$5,270,000	79% (\$4,144,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Anderson-Pendleton 100 kV	Rebuild line with higher-capacity conductor	\$4,244,000	79% (\$3,338,000)
Murdock-Napavine 100 kV	Rebuild line with higher-capacity conductor	\$10,995,000	79% (\$8,646,000)
Rmec-Christopher 100 kV line	Rebuild line with higher-capacity conductor	\$2,709,000	79% (\$2,130,000)
Wrangell-Devers 100 kV	Reconductor line	\$4,723,000	79% (\$3,714,000)
Shiprock 2 Substation-Makahoy 100 kV	Reconductor line	\$1,623,000	79% (\$1,276,000)
	Total allocated to project		\$26,545,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.